THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LONGIDO DISTRICT

FORM THREE ANNUAL EXAMINATION
013 GEOGRAPHY
MARKING SCHEME NOVEMBER 2025

SECTION A (16 MARKS)

1.

(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)	(x)
A	D	A	E	B	D	C	B	E	D

10 @01%= 10%

2.

(i)	(ii)	(iii)	(iv)	(v)	(vi)
B	G	E	D	I	H

06 @)1% = 06%

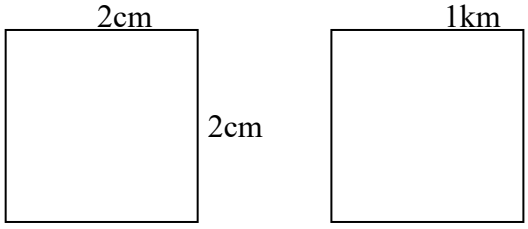
SECTION B (54 MARKS)

3. (a) Latitudes is 4 50 S and longitudes 29 40

(b) Distance in cm is 30 cm **……..…01%**
Map scale 1:50,000
Map scale 1cm = ½ km
1 cm = ½ km**……..……...01%**
30 cm = X
30 cm x ½km=15Km

Distance of the road from grid reference 020580 to 910606 =15Km **…..01%**

(c) Full squares 180**……..……01%**
Total Squares 180
Map scale 1:50,000
1cm:1/2 km
2x1cm:1/2kmx2
2cm:1km.…**……01%**



Area= Length x Width
Area = 1 km x 1km

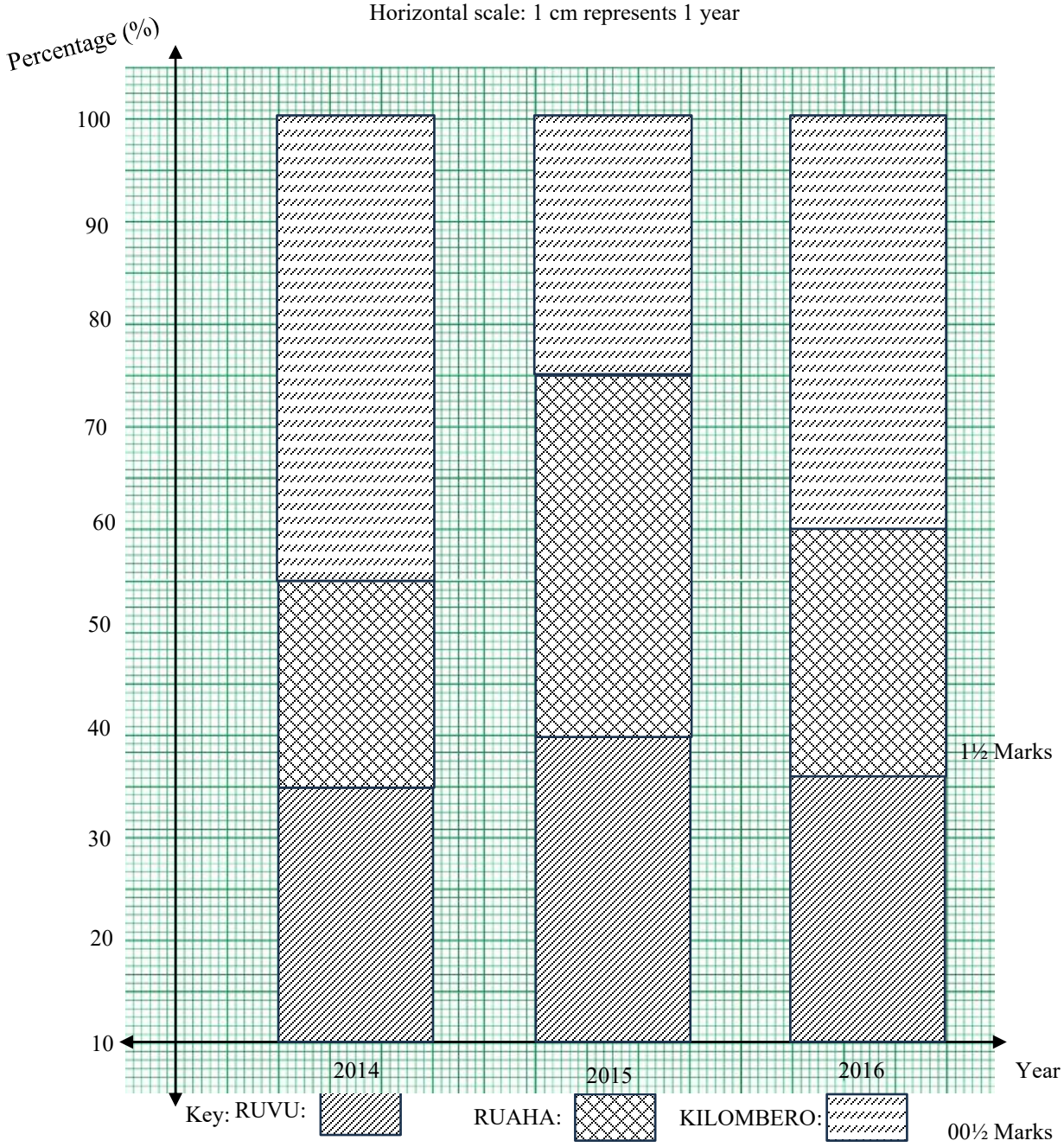
Area = 1 km²
180 square x 1 km²=180 km²
Area of the Burka Estate is 180 km²**…….01%**

(d) Three economic activities can be conducted on the mapped area are
(i) Transportation and navigation 3@01%
(ii) Agriculture …..**.....03%**
(iii)Trade
(iv)Fishing

4. (a).

COMPOUND BAR GRAPH SHOWING WATER FLUCTUATION IN RUAHA, RUVU AND KILOMBERO FROM 2014 TO 2016 00½ Marks

Vertical scale: 1 cm represents 10% 00½ Marks
Horizontal scale: 1 cm represents 1 year



(b).
(i) Evaporation

(ii) lack of rainfall
(iii)water supply use
(iv)water flow requirements

(c). key advantages
(i) highlights proportional contributions
(ii) efficiently compares categories
(iii)support trend analysis
(iv)Reduces data clutter
(v) Clearer interpretation
(vi)easy to understand

5. (a). Hypothesis

(b). **Provides Focus and Direction**

- A hypothesis helps narrow the scope of a study, ensuring that research efforts are targeted and productive.
- It acts like a compass, guiding the researcher by defining the specific problem and the variables to investigate.

Structures the Research Process

- It serves as a framework for data collection and analysis, helping researchers select appropriate methods, tools, and statistical tests.
- By formulating a hypothesis, researchers can organize their experiments and observations systematically.

Links Theory and Evidence

- A hypothesis connects theoretical assumptions to scientific observations, bridging existing knowledge and new investigations.
- It is not a mere guess but an informed suggestion derived from existing knowledge, previous research, or observations.

Enables Testability and Verifiability

- A well-formulated hypothesis must be testable and falsifiable, meaning it can be proven wrong by empirical evidence.
- This allows for the validation or refinement of the hypothesis, which is crucial for advancing scientific knowledge.

Facilitates the Advancement of Knowledge

- Through the rigorous testing and refinement of hypotheses, scientists can uncover new insights, develop theories, and solve complex problems.
- This continuous cycle of hypothesis-driven research leads to the creation of new, dependable, and generally applicable knowledge

(c). 1. **Testable**

A hypothesis must be verifiable or falsifiable, meaning that you can collect observable data through scientific methods to either support or disprove the statement.

It cannot be a mere guess but a predictive statement of an outcome that can be investigated.

2. **Specific**

A specific hypothesis clearly defines the variables being studied and the relationship between them.

It should articulate what is being assessed, who or what variables are involved, and the expected outcome.

3. Clear and Concise

The language used in a hypothesis should be easily comprehensible and precise, avoiding ambiguity.

It must be written in a simple and clear manner to ensure that the target audience can understand its meaning without confusion.

(d). **Ambiguous or Untestable Hypothesis**:

A hypothesis needs to be clear, precise, and capable of being tested through empirical data collection. If the hypothesis is poorly formulated, it becomes difficult to design a study with appropriate methods and variables, making it nearly impossible to test and draw valid conclusions.

Lack of Suitable Research Methods or Resources:

Even a well-formed hypothesis can present problems if the researcher lacks the necessary methods, tools, or financial resources to test it. This can include difficulties in accessing the right participant population, collecting accurate data, or using appropriate analytical techniques.

6. (a) Gross error**…….01%**

(b). **Defined Survey Objectives and Target Population:**

- The team needs a clear understanding of why they are surveying and who they are surveying.
- Discrepancies can arise if the survey is performed on the wrong group of people or for the wrong purpose, leading to incorrect assumptions or measurements.

Appropriate Measurement Tools and Methods:

- Using incorrect or improperly calibrated measurement equipment can introduce errors.
- The team must ensure they use the correct tools and understand how to apply them effectively for the specific terrain and required accuracy.

Precise Fieldwork Procedures and Alignment:

- A common issue is a "chainage error," where a measuring chain or tape is not kept straight between points.
- The team needs to ensure that the poles are erected with proper alignment and straight lines, maintaining correct distances between them.

Quality Control and Pre-Testing:

- Regular checks of the survey equipment and the work being performed by the field team are essential.
- Pre-testing the survey plan and methods can identify potential issues before actual pole erection begins, helping to prevent discrepancies.

(c). (i). Being clear,
(ii). consistent,
(iii). timely with all entries,
(iv). using proper headings and units,
(v). Making comprehensive notes with detailed sketches to avoid misinterpretation. Mistakes should be corrected by crossing them out, not overwriting, with canceled pages clearly marked and referenced, ensuring the field book is kept clean and organized for easy plotting by other draftsmen.

7. (a) 1. Plate Tectonics

Plate Movement:

The Earth's outer crust is made of large tectonic plates that are constantly in motion due to convection in the mantle.

•**Stress Accumulation:**

Plates can get stuck due to friction as they try to move, causing stress to build up within the rocks.

- **Sudden Release:**

When the accumulated stress exceeds the strength of the rocks, the rocks suddenly break or slip along a fault plane, releasing stored energy in the form of seismic waves that cause the ground to shake.

2. Volcanic Activity

- **Magma Movement:**

The movement of magma beneath the Earth's surface can create stress on surrounding rocks.

- **Rock Fracturing:**

This stress can cause the rocks to fracture, which in turn generates seismic waves and results in a volcanic earthquake.

3. **Isostatic Adjustments**

- **Vertical Movement:** This process involves the vertical sliding of rock bodies past each other.
- **Cause:** Isostatic adjustments can occur due to shifts in the Earth's crust, such as postglacial rebound (the rising of land after an ice sheet melts).

(b). magnitude scale

(c). 1. Kilimanjaro-Arusha-Dodoma Fault Zone

- **Location:**

This belt runs roughly from Mount Kilimanjaro southwards through Arusha and Dodoma.

- **Characteristics:**

It is one of the major geological fault systems within Tanzania and has experienced significant seismic activity. For instance, a 5.1 magnitude earthquake struck Dodoma in 2014.

- **Example:**

The 2014 Dodoma earthquake is an example of the seismic events occurring in this zone.

2. Kigoma-Mbeya Rift Segment (including the Lake Tanganyika area)

- **Location:**

This belt stretches from Kigoma in the west to Mbeya in the southwest, with Lake Tanganyika marking the western boundary.

- **Characteristics:**

This region is particularly active, with frequent and strong earthquakes. It is part of the Tanganyika-Rukwa-Malawi rift segment of the East African Rift System.

- **Example:**

The region around Lake Tanganyika has recorded some of Tanzania's strongest earthquakes, including a magnitude 7.4 event in 1910 and a 6.8 magnitude earthquake in 2005

(d) 1. Drop, Cover, and Hold On

- **Drop**: to your hands and knees to protect your body from being knocked down.
- **Cover**: your head and neck with your arms and crawl under a sturdy table or desk for shelter.
- **Hold on**: to your shelter with one hand until the shaking stops. If there's no shelter, crouch against an interior wall and cover your head and neck.

2. Secure Your Home

- **Anchor**: heavy furniture, water heaters, and tall objects to walls to prevent them from falling and causing injury.
- **Secure**: light fixtures and other hanging objects.

- **Consider**: structural improvements to your home, like securing it to its foundation.

3. Create a Preparedness Plan
   - **Identify**: safe spots in each room, such as under sturdy tables.
   - **Assemble**: a disaster supply kit with essentials like water, non-perishable food, a flashlight, and sturdy shoes.
   - **Practice**: earthquake drills with your family and establish a communication plan

8. (a). Oblique photograph

(b) Transportation

- **Boat-based transport:**
  Rivers are natural pathways for transporting goods and people, which can form the basis of a local economy.
- **Port and logistics hubs:**
  Areas around rivers often develop ports and logistics centers to manage the movement of cargo.

Resource Extraction and Agriculture

- **Fishing:**
  The river can be a source of fish for local markets or larger food industries.
- **Agriculture:**
  Rivers provide water for irrigation, enabling diverse farming activities in the surrounding land.

Energy Production

- **Hydroelectric power:** Dams can be built on rivers to generate electricity, a significant economic resource.

Tourism and Recreation

- **Water sports:**
  Activities such as boating, rafting, kayaking, and swimming attract tourists, creating jobs in guiding and equipment rental.
- **Scenic tourism:**
  The natural beauty of a river can draw visitors, supporting hotels, restaurants, and souvenir shops.

Industrial and Urban Development

- **Industrial processes:**
  Many industries rely on rivers for water for cooling or as a source of raw materials, leading to industrial development along riverbanks.
- **Waste disposal:**
  Rivers can be used for the disposal of industrial and urban wastewater, although this has environmental consequences.

Other Potential Activities

- **Mining:**
  In some cases, minerals or other resources might be extracted from the riverbed or surrounding alluvial deposits.
- **Landscaping and real estate:**
  Areas with rivers can be attractive for housing and commercial development, creating value for land.

(c) linear settlement

They have been established along the road and river banks

(d) The presence of

(i) rivers

(ii) mountains

SECTION C (30 MARKS)

9. Mining is encouraged despite environmental side effects primarily due to its essential role in providing raw materials for modern society, driving economic growth through job creation and revenue generation, and offering opportunities for local development. The high demand for minerals and metals, combined with the lack of robust implementation of environmental regulations and the limited capacity of some governments to negotiate fair terms, often outweighs the negative impacts, leading to its continued promotion.

   Good introduction…….01.5%

   1. Essential for Modern Society:
   Minerals and metals are crucial for manufacturing countless products and for building infrastructure, making mining a fundamental requirement for modern economies and living.
   2. Economic Growth and Employment:
   Mining creates jobs and generates revenue, which can boost national and local economies, provide funds for development projects, and offer significant opportunities for resource-rich countries.
   3. Demand for Minerals and Metals:
   There is continuous global demand for the raw materials extracted through mining, necessitating a stable rate of new mine development to ensure a steady supply for industries and consumers.
   4. Lack of Regulatory Capacity:
   Many resource-rich nations lack the capacity to effectively enforce environmental and social regulations, leading to poor management of mining projects and a lagging implementation of assessment and closure rules.
   5. Insufficient Legal Frameworks:
   Legal and contractual frameworks governing mining operations may be designed with insufficient consideration for environmental sustainability and community well-being, often failing to secure the full economic benefits for the country.
   6. Interconnectedness with Global Markets:
   Mining is tied to global economic changes, and the need for resources on a global scale can drive mining operations even when there are known local environmental and social costs, especially if alternative solutions are not yet viable.
   Any good conclusion ……**01.5%**

10. Long distances to a water source negatively affect girl children by reducing their time for education, contributing to physical strain and health issues from carrying heavy loads, and exposing them to increased risk of physical harm and social neglect. The extensive physical demands and lost time limit their academic performance, curtail childhood development, and can trap them in a cycle of poverty. Any good introduction…….02%**.**

    Here are six specific ways distance to water affects the girl child:

    1. Reduced Educational Opportunities:
       Girls often spend significant time fetching water, leaving less time and energy to focus on their studies, which can lead to lower academic performance and increased school absenteeism.
    2. Physical Strain and Health Problems:
       Carrying heavy water containers over long distances can cause physical pain, chronic fatigue, and long-term health issues, including spinal disorders and cervical compression syndromes.

3. Increased Risk of Accidents and Harm:
   The long walks to water sources, especially in potentially unsafe rural environments, increase a girl's risk of accidents, injury, and exploitation.
4. Limited Social Development:
   Time spent on water collection reduces opportunities for play and social interaction, which are vital for a child's overall social and emotional development.
5. Cycle of Poverty:
   When girls are burdened with water fetching duties, they are often unable to pursue education or engage in income-generating activities, trapping them and their families in a cycle of poverty.
6. Mental Health Impacts:
   The constant physical and time burden of water collection, coupled with institutional injustices and the demands of a household without accessible water, can negatively affect a girl child's mental health.
   Six points 6 @ 2%.....12%
   Any good conclusion ……**01%**

11. Agriculture serves as a national economic backbone by providing substantial contributions to the Gross Domestic Product (GDP) and export earnings, supplying food and fodder for the population, generating raw materials for industries, creating widespread employment opportunities, supporting rural livelihoods and stability, and contributing to foreign exchange earnings through agricultural exports.
    introduction.....2marks

Here are six (6) key points to discuss how agriculture acts as a backbone of the economy:

1. **Significant GDP Contribution:**
   Agriculture is often a large and significant contributor to a country's overall GDP, forming a foundational economic sector. For example, in Tanzania, it contributes a substantial percentage to the GDP.
2. **Source of Employment:**
   The sector provides a large number of employment opportunities, particularly for rural populations, making it a primary source of livelihood and a key driver of social stability.
3. **Raw Material Supply for Industries:**
   Agriculture provides essential raw materials, such as cotton, timber, and food crops, to a wide range of industries, supporting their manufacturing processes and enabling further economic diversification.
4. **Food and Nutrition Security:**
   By supplying the nation's food and animal fodder, the agricultural sector ensures food security for the population and contributes to nutrition security, which is vital for public health and well-being.
5. **Major Source of Exports and Foreign Exchange:**
   Agricultural commodities are a significant source of export revenue for many countries, providing essential foreign exchange earnings that strengthen the national currency and economic standing.
6. **Foundation for Rural Livelihoods:**
   In rural areas, agriculture is the cornerstone of the economy, directly supporting the livelihoods of the majority of rural inhabitants and serving as a stable economic base for these communities.
   Any six points 6@ 2marks .....12 marks

   Any good introduction……..……01%